

Solution & System Concept Document

2.1 Solution Overview

Saarthi is a **mobile-first, offline-capable AI heritage guide** that provides accurate, contextual, personalized, and multilingual information to tourists while they are physically exploring heritage sites. It combines:

- **Location-aware content** (GPS + geofencing + optional AR) to trigger explanations for specific structures, sculptures, inscriptions, and viewpoints.
- **Verified heritage knowledge** (ASI/UNESCO/state sources + expert-reviewed content) with clear source citations and "original vs. restored vs. reconstructed" labels.
- **Multilingual AI** (Hindi + major Indian languages + key foreign languages) for text, audio, and simple speech interaction.
- **On-site navigation** (entrances, routes, facilities, accessible paths) and real-time context (crowd levels, closures, weather-aware suggestions).
- **Offline-first design** (downloadable site packs: maps, audio, text, QR/AR content) for low-connectivity monuments.

2.2 Problem → Solution Mapping

Identified Problem	Saarthi Solution
Fragmented information — history, tickets, maps, facilities, accessibility across multiple platforms	Unified heritage platform — single app integrating verified content, navigation, facilities, tickets, and accessibility for each site
Unreliable information — insufficient official content, commercial misinformation	Verified ASI/UNESCO/state/expert content with source citations, provenance labels, and "evidence vs. interpretation" markers
Poor contextual interpretation — basic facts without historical, architectural, cultural depth	Location-specific storytelling — structure-level narratives covering historical phases, architecture, cultural meaning, community perspectives, and authenticity labels
Navigation gaps — poor signage, missing facility info, no internal routing	GPS + geofencing + QR + interactive maps — zone-level and structure-level triggering with facility navigation and accessible routes

Language barriers — inconsistent multilingual support	Multilingual text/audio/voice — Hindi, major Indian languages, key foreign languages with human-reviewed translations + TTS
Limited personalization — one-size-fits-all content	AI-generated micro-itineraries — routes tailored to time, budget, interests, mobility, age, weather, traffic, and crowd conditions
Connectivity problems — weak internet at heritage sites	Offline site packs — downloadable maps, audio, text, QR content, accessibility info, and selected AR/3D assets
Accessibility gaps — missing ramps, inaccessible info	Accessible routes + audio/captions — step-free paths, screen-reader design, simple-language mode, optional sign-language videos
Guide verification — unverified informal guides	QR-based guide authentication — verified guide profiles with license status and credential display
Lesser-known sites — poor visibility and coverage	AI discovery & recommendations — personalized suggestions for nearby lesser-known monuments based on interests and itinerary
Crowd issues — overcrowding at popular sites	Crowd-aware routing/slot information — real-time crowd levels, dynamic routing, and conservation alerts
Local economic exclusion — artisans and communities left out	Artisan/homestay/guide marketplace — verified local experiences, workshops, and cultural interactions integrated into the platform

2.3 Core Features

A. Verified Heritage Knowledge

- Historical phases and timelines
- Architectural analysis and style identification
- Cultural context and significance
- Community narratives and oral histories (labeled as such)
- Original / restored / reconstructed / interpretive labels
- Source citations to ASI, UNESCO, state archaeology, and expert-reviewed records
- Multiple perspectives where relevant (archaeological evidence, local traditions, alternative interpretations)

B. Location-Aware Exploration

- GPS-based zone triggering (3–10 m accuracy for coarse zones)

- Geofencing for structure-level content activation
- QR code scanning for precise point-of-interest information
- Optional AR overlays for immersive interpretation
- Structure-level information for specific buildings, sculptures, inscriptions, galleries, and viewpoints

C. Multilingual Voice-First Experience

- Hindi and major Indian languages
- Key foreign languages (English, French, Spanish, German, Japanese, Chinese, etc.)
- Audio narration with human-reviewed translations + TTS
- Speech interaction (speech-to-text for queries)
- Captions for all audio content
- Simple-language mode for younger or less specialized visitors
- Screen-reader friendly design

D. AI Journey Planner

- Budget-aware planning
- Time-constrained itineraries (30–45 min, 2–3 hr, half-day)
- Interest-based routing (architecture, religion, photography, history)
- Mobility-adaptive paths (wheelchair, elderly, families with children)
- Weather-aware suggestions (shade, water points, best visit times)
- Traffic and transport integration
- Crowd-condition-aware scheduling
- Personalized multi-site routes

E. Offline Heritage Packs

- Downloadable site maps (vector tiles)
- Offline audio tours
- Full text content
- QR-linked content cache
- Accessibility information

- Selected AR/3D assets with LOD optimization
- Facility maps and route information

F. AR/3D Heritage

- Historical reconstruction overlays (e.g., showing a ruin as it originally appeared)
- Architectural visualization and structural explanations
- Interactive storytelling triggered by camera recognition
- Gamified exploration elements for engagement
- Graceful degradation to 2D/text/audio for non-AR devices

G. Visitor Management

- Real-time crowd level information
- Dynamic routing to avoid congestion
- Closure and restriction notifications
- Conservation work alerts
- Visitor analytics for site managers (aggregated, anonymized)

H. Local Marketplace

- Verified local guides with credential display
- Artisan profiles and product showcases
- Homestay listings near heritage clusters
- Cultural experiences and workshops
- Community-led heritage walks

2.4 How Saarthi Works

User Journey Flow:

Discover → Plan → Download Site Pack → Reach Site → Explore → Identify/Scan → Learn → Personalize → Discover Local Experiences → Explore Responsibly

Detailed Flow:

1. **Discover** — User browses or searches for heritage sites; AI recommends lesser-known destinations based on interests and location.

2. **Plan** — AI Journey Planner generates a personalized itinerary considering time, budget, interests, mobility, weather, traffic, and crowds.
3. **Download Site Pack** — User downloads the offline site pack (maps, audio, text, QR/AR content) over Wi-Fi before visiting.
4. **Reach Site** — GPS detects arrival at the heritage site; app switches to on-site mode.
5. **Explore** — Location-aware content triggers automatically as the user walks through zones; navigation guides to entrances, key stops, facilities, and viewpoints.
6. **Identify/Scan** — User scans QR codes or points camera at structures for precise information; optional AR overlays activate.
7. **Learn** — Verified, contextual, multilingual narratives are delivered via text, audio, or voice interaction; source citations and authenticity labels are displayed.
8. **Personalize** — User adjusts depth, language, pace, and interests in real time; AI adapts the remaining route.
9. **Discover Local Experiences** — App surfaces verified local guides, artisans, homestays, and cultural workshops nearby.
10. **Explore Responsibly** — Conservation alerts, crowd-aware routing, and sustainable tourism guidance help minimize visitor impact.

2.5 Existing Technology Foundation

Saarthi does **not** require inventing new technology. It integrates and orchestrates existing, proven technologies:

Technology	Role in Saarthi	Maturity
AI / RAG (Retrieval-Augmented Generation)	Source-grounded heritage explanations with citations	Production-ready
LLMs (Large Language Models)	Natural-language interaction, summarization, translation	Production-ready
GPS	Location detection and zone triggering	Ubiquitous in smartphones
ARCore / ARKit	Augmented reality overlays and tracking	1,200+ Android models supported by 2026
Maps APIs (Google Maps, Mapbox)	Navigation, routing, offline tiles	Mature, well-documented

Technology	Role in Saarthi	Maturity
QR codes	Physical-to-digital connection at specific points	Universal
Translation APIs (Cloud NMT, Sarvam, etc.)	Multilingual content generation	Production-ready for major languages
TTS / STT	Text-to-speech narration and speech-to-text queries	Mature
Cloud infrastructure (AWS, Azure, GCP)	Backend hosting, CDN, storage	Mature; India regions available
3D / Photogrammetry	Digital twins and AR assets	Established workflows
Computer vision	Object/structure recognition for AR triggers	Production-ready

UNESCO alignment: UNESCO itself identifies technologies such as 3D scanning, modelling, digital preservation, AI, and AR/VR as tools that can improve access to and preservation of cultural heritage. Saarthi's technology choices are fully consistent with this framework.

2.6 Existing Ecosystem Integration

Saarthi **builds on top of** existing systems rather than replacing them:

Existing System	Saarthi Integration
ASI / UNESCO	Ingests verified knowledge — monument data, conservation status, historical records — as the authoritative content layer
Incredible India	Leverages tourism and destination information; the Incredible India Content Hub already aggregates assets from the Ministry of Tourism, Ministry of Culture, and ASI
Maps (Google / Mapbox)	Uses existing navigation and routing infrastructure for on-site and last-mile guidance
Audio guides (Audio Odigos)	Complements and extends audio interpretation with location-triggering, personalization, and multilingual expansion
QR infrastructure	Uses QR codes as physical-to-digital bridges already deployed at many monuments

Existing System	Saarthi Integration
AR/3D projects	Integrates existing 3D scans and AR assets (e.g., from government heritage digitization projects) into the unified platform
Saarthi's role	Unified intelligent on-site experience — the integration layer that combines all of the above into a single, personalized, offline-capable, location-aware guide